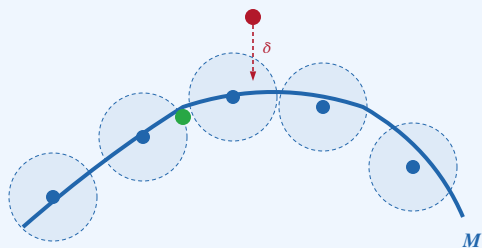


(1) QTS (ours)

Prototype matching · input space



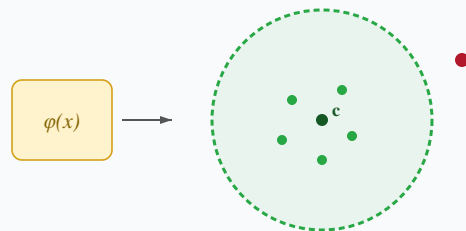
M prototypes cover normal manifold

$$\text{Score} = \min_k \|x - (-w_k)\|^2$$

✓ Coverage + separation guarantees

(2) Deep SVDD

One-class boundary · feature space



Single center c in learned ϕ -space

$$\text{Score} = \|\phi(x) - c\|^2$$

✗ Single center; opaque feature space

(3) Autoencoder

Reconstruction · bottleneck



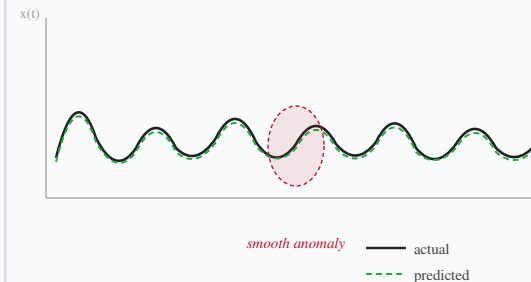
Anomaly reconstructed nearly perfectly

$$\text{Score} = \|x - g(f(x))\|^2$$

✗ No separation guarantee (Prop 3)

(4) TranAD

Forecasting · self-attention



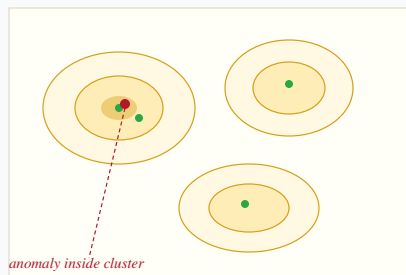
Predict next; residual = anomaly score

$$\text{Score} = \|x - x_{\text{pred}}\|^2$$

✗ Smooth anomalies are predictable

(5) DAGMM

Density estimation · GMM in latent



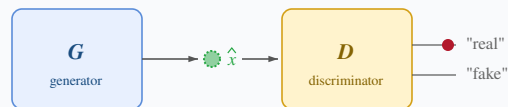
Fit GMM to latent embeddings

$$\text{Score} = -\log p_{\text{GMM}}(z)$$

✗ Wrong family $\rightarrow$ high p for anomaly

(6) BeatGAN

Adversarial reconstruction



G covers anomaly manifold;
 D fooled, $\|x - \hat{x}\| \approx 0$

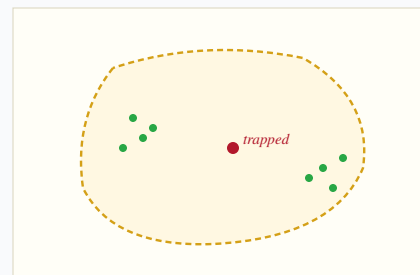
G learns to reconstruct, D ranks

$$\text{Score} = \alpha \cdot \|x - G(x)\|^2 + \beta \cdot D\text{-loss}$$

✗ Generator over-generalises

(7) OC-SVM

Kernel boundary · classical



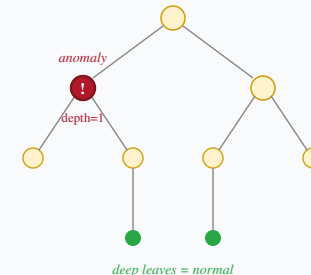
RBF kernel hypersphere in feature

$$\text{Score} = \langle w, \phi(x) \rangle - \rho$$

✗ Single boundary; shallow features

(8) Isolation Forest

Tree partitioning · classical



Random splits isolate anomalies fast

$$\text{Score} \propto 1 / E[\text{path-depth}(x)]$$

✗ Axis-aligned splits ignore morphology